

Day 20 - Lecture 20.10: Conditional Probability

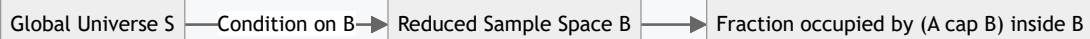
 Python 3.10+

 notes.pdf

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1. Overview & Sample Space Reduction

Conditional Probability measures the likelihood of an event A occurring given that another event B has already occurred ($P(A | B)$). Geometrically, conditioning on event B shrinks the relevant universe from the entire sample space $\mathcal{S}$ down exclusively to the subset B .



2. Mathematical Formalism

Formal Definition:

For any two events A and B where $P(B) > 0$:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

In a finite uniform discrete sample space with cardinality counting:

$$P(A | B) = \frac{|A \cap B|}{|B|}$$

Kolmogorov Axiom Consistency:

Conditional probability is a rigorous probability measure satisfying all three Kolmogorov axioms over the restricted universe B :

1. $P(A \mid B) \geq 0$
 2. $P(B \mid B) = 1.0$
 3. $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B)$ if $A_1 \cap A_2 = \emptyset$.
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3. Real-World Demonstrative Example

A medical dataset contains 1,000 patient records classified by smoking status and cardiovascular disease (CVD):

- Smoker (B): 300 patients
- Non-Smoker (B'): 700 patients
- Smoker AND Has CVD ($A \cap B$): 90 patients
- Non-Smoker AND Has CVD ($A \cap B'$): 35 patients

What is the probability that a randomly chosen patient has CVD given that they are a smoker?

$$\boxed{P(\text{CVD} \mid \text{Smoker}) = \frac{P(\text{CVD} \cap \text{Smoker})}{P(\text{Smoker})} = \frac{90 / 1000}{300 / 1000} = \frac{90}{300} = 0.30 \text{ quad } (30\%)}$$

What is the baseline unconditional probability of CVD across the general population?

$$\boxed{P(\text{CVD}) = \frac{90 + 35}{1000} = \frac{125}{1000} = 0.125 \text{ quad } (12.5\%)}$$

Because $P(\text{CVD} \mid \text{Smoker}) = 0.30 \neq P(\text{CVD}) = 0.125$, smoking and cardiovascular disease are statistically dependent.

4. Python Implementation

```
import pandas as pd

# Contingency table representation
data = {
    'CVD_Yes': [90, 35],
    'CVD_No': [210, 665]
}
df = pd.DataFrame(data, index=['Smoker', 'NonSmoker'])

total = df.values.sum()
p_smoker = df.loc['Smoker'].sum() / total
p_cvd_and_smoker = df.loc['Smoker', 'CVD_Yes'] / total

# Conditional Probability  $P(\text{CVD} \mid \text{Smoker})$ 
p_cvd_given_smoker = p_cvd_and_smoker / p_smoker

print("Contingency Table:\n", df)
print(f"\nP(Smoker):                {p_smoker:.4f}")
print(f"P(CVD and Smoker):            {p_cvd_and_smoker:.4f}")
print(f"P(CVD | Smoker):              {p_cvd_given_smoker:.4f} (Exact: 90/300 = {90/300})")
```

5. Key Takeaways & Interview Points

- **Universe Reduction:** Conditioning on B changes the denominator from $|\mathcal{S}|$ to $|B|$.
- **Conditioning vs Intersection:** $P(A \cap B)$ is the probability of both happening in the global universe; $P(A \mid B)$ is the probability of A happening once we already know B is true.